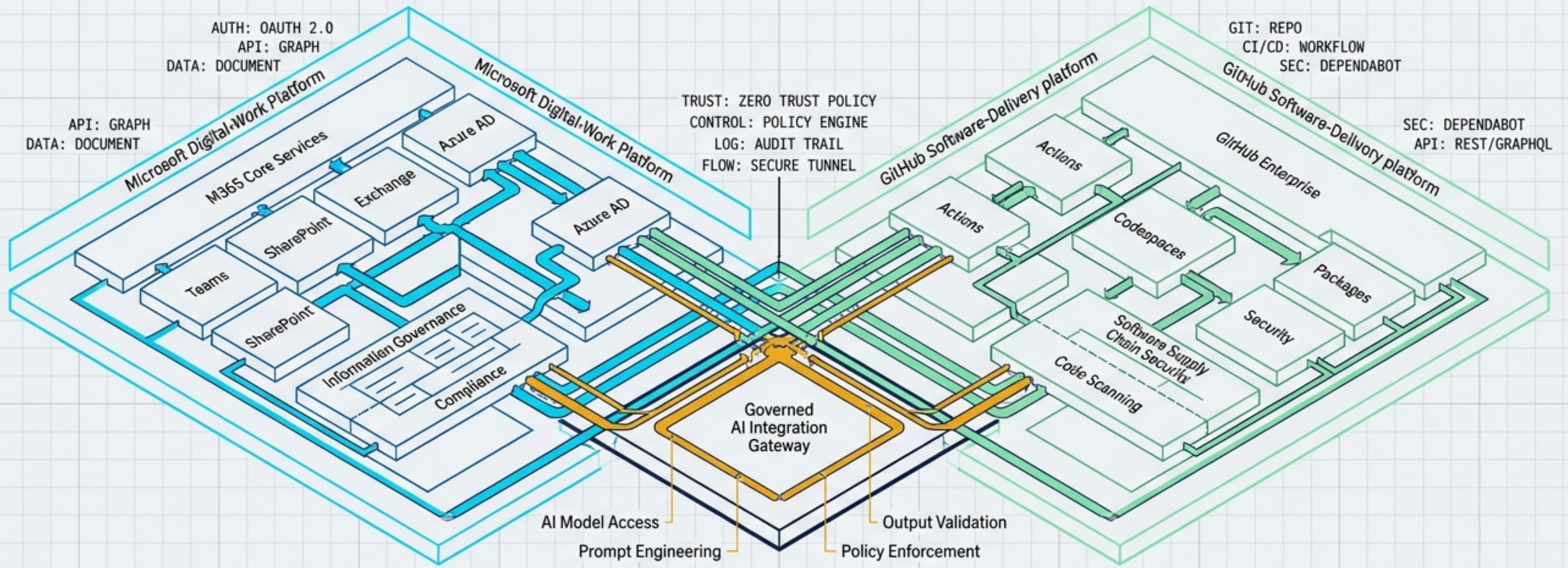


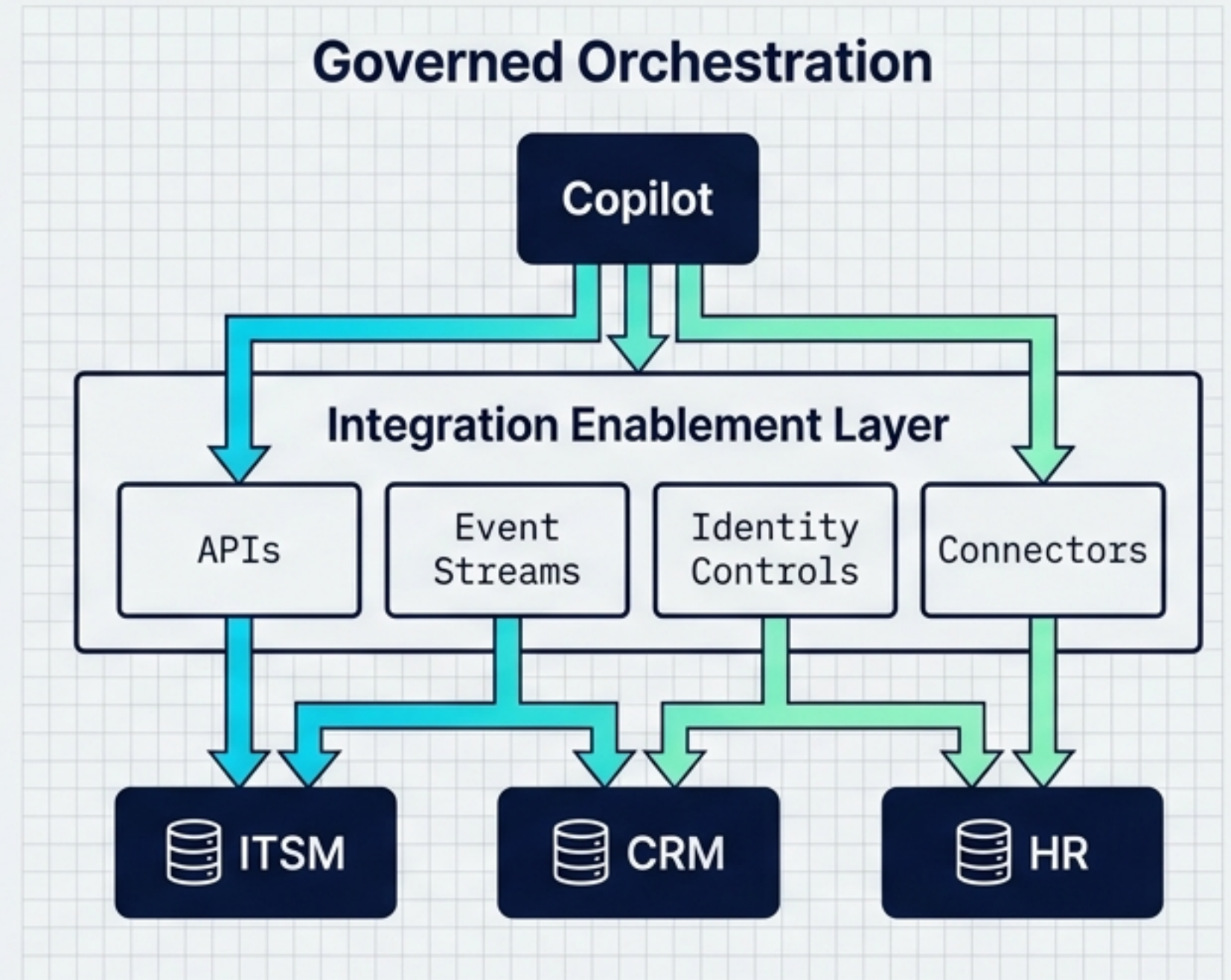
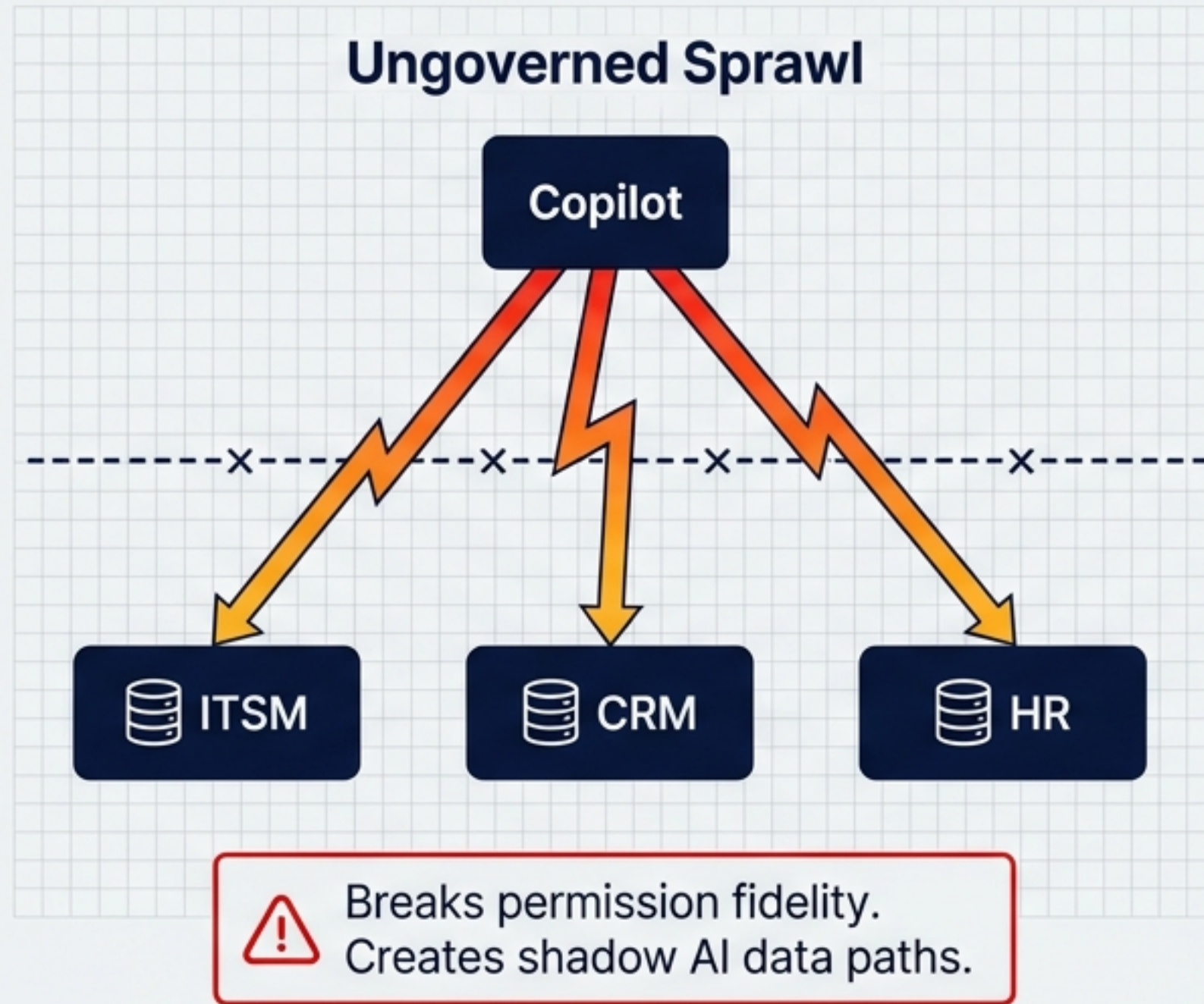


The Blueprint for Governed Enterprise AI

Architectural Framework: Integrating AI across Microsoft 365 and GitHub without bypassing enterprise controls.

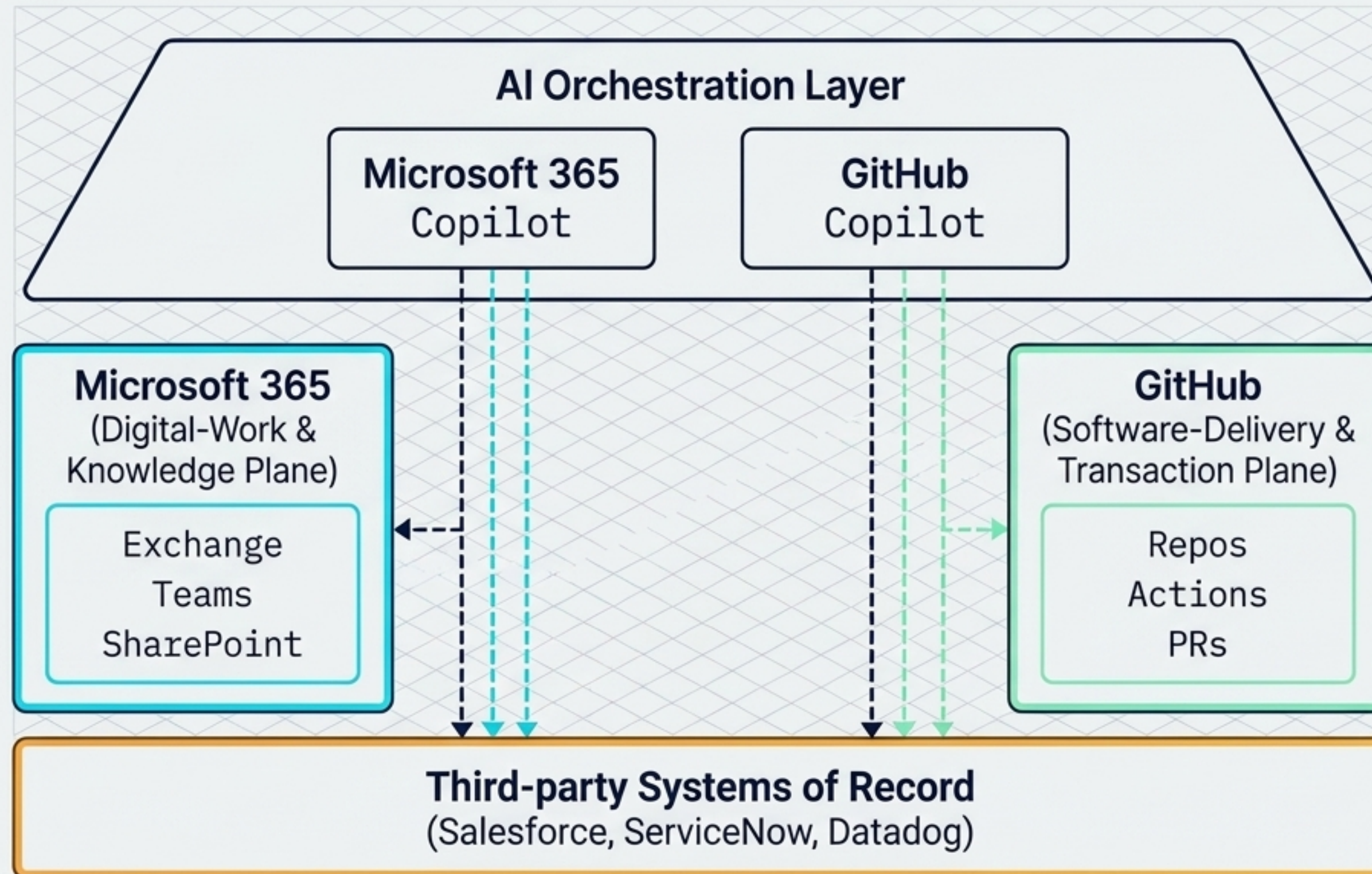
Prepared for Enterprise Architecture & Security Leadership.

AI is a Governed Consumer, Not an Integration Bypass



Copilot is a user-facing intelligence layer. It must orchestrate existing, authorized integrations rather than replacing them. Without this, enterprise permission fidelity shatters.

The Conceptual View: Two Platforms, One Strategy

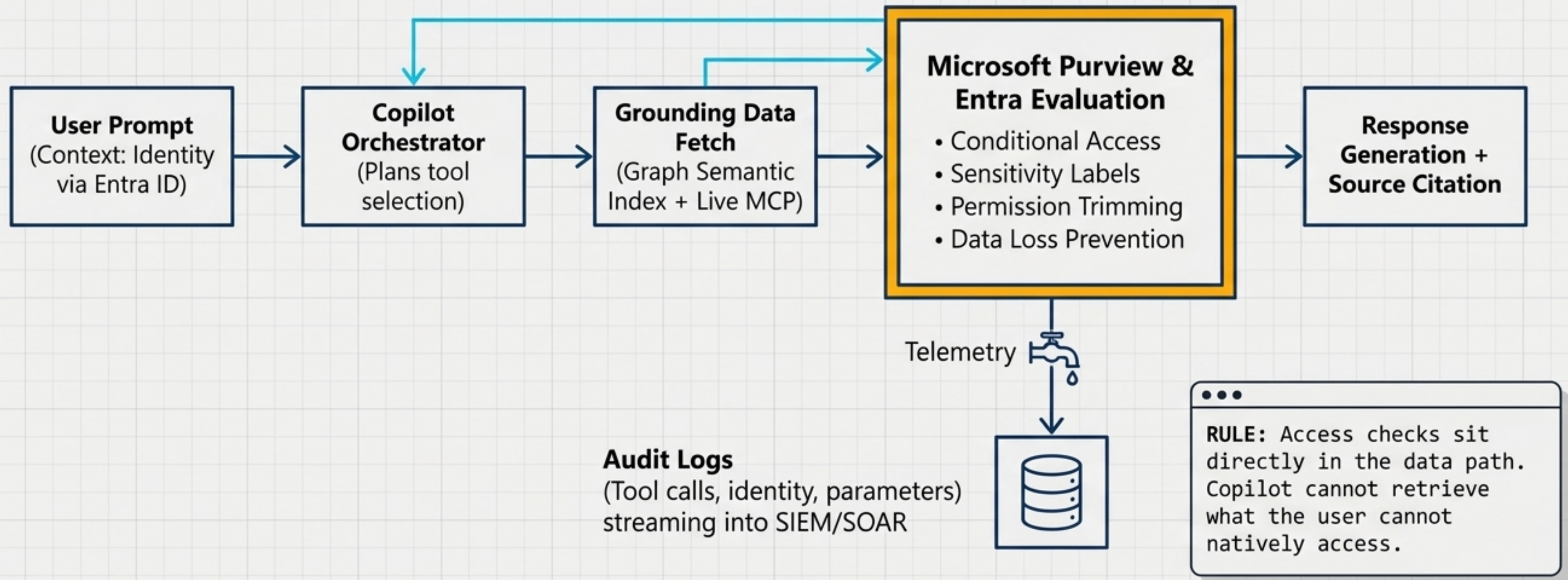


Key Takeaway

Each platform maintains its own native AI orchestration layer.

Third-party data, identity, and actions must cross the boundary safely using governed APIs, Model Context Protocol (MCP), and federated identity.

The AI Data Path and Governance Chokepoint



Mapping the Integration Surfaces

Microsoft 365 Surface	GitHub Surface	The Shared Standard
• Graph API: Read/write resources • Graph Data Connect: Bulk analytics • Copilot Studio: Business agents • Copilot APIs: Embedded intelligence	• REST/GraphQL APIs: Operational actions • Webhooks/Actions: Event-driven automation • GitHub Apps: Durable server-to-server	Model Context Protocol (MCP) servers Primary extensibility mechanism for Copilot across both platforms.

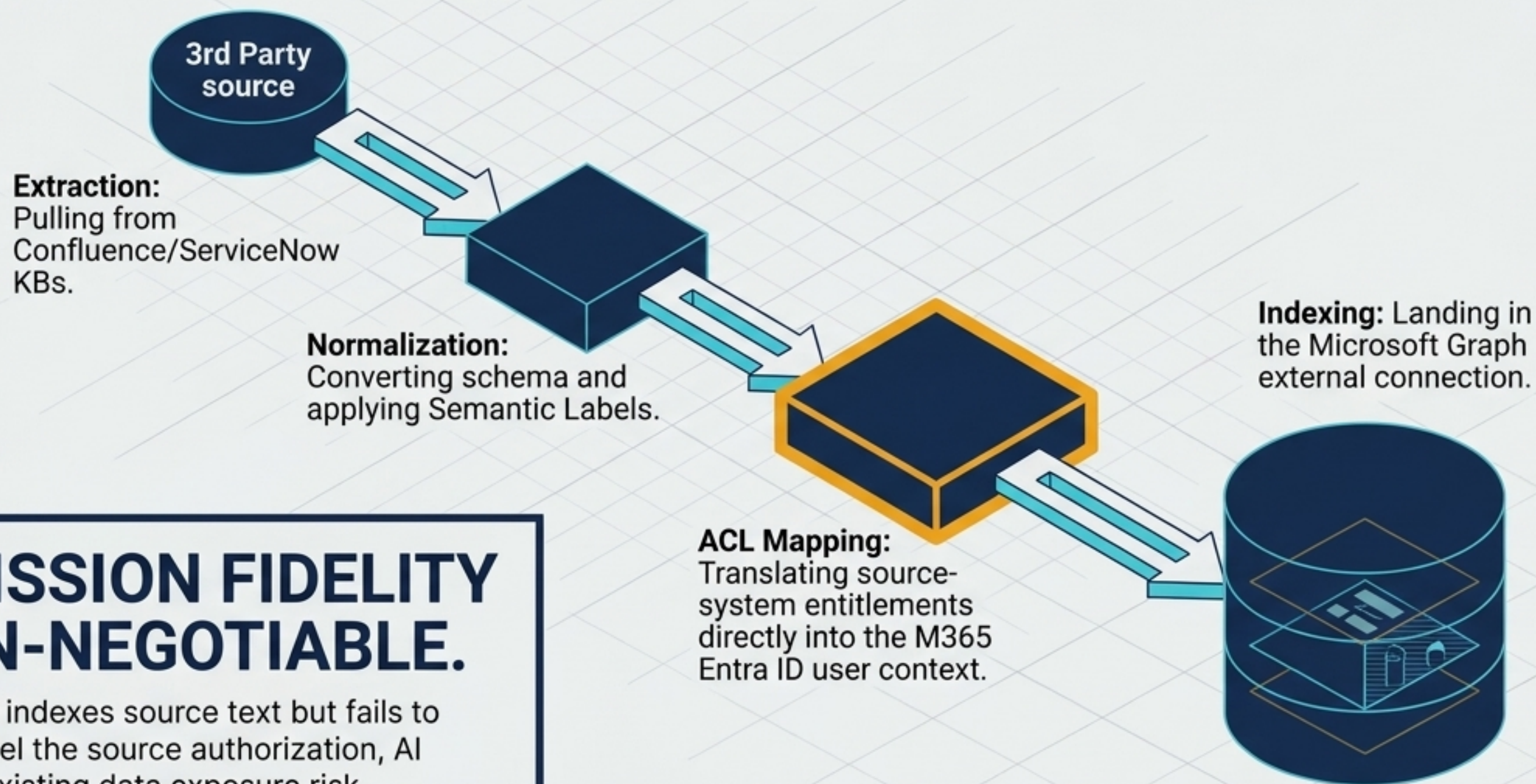
Summary: The transition from proprietary extensions to the open MCP standard bridges the extensibility gap between the work and software-delivery domains.

The Great Divide: Synced vs. Federated Connectors

Synced Connectors (Indexed)	Federated Connectors (MCP)
Mechanism: Data extracted, ACL-mapped, and copied into Microsoft Graph Semantic Index.	Mechanism: Fetched live at query-time via MCP; nothing persists in Graph.
Primary Use: Broad enterprise knowledge retrieval, durable search.	Primary Use: Real-time facts, sensitive data, strict regulatory boundaries.
Speed of Change: Static or slow-changing.	Speed of Change: Highly dynamic.
Capabilities: Read-only.	Capabilities: Read and governed Write actions (via tool calling).

Actionable Insight: Never index highly dynamic or sensitive transactional data. Federation (MCP) preserves the authoritative source.

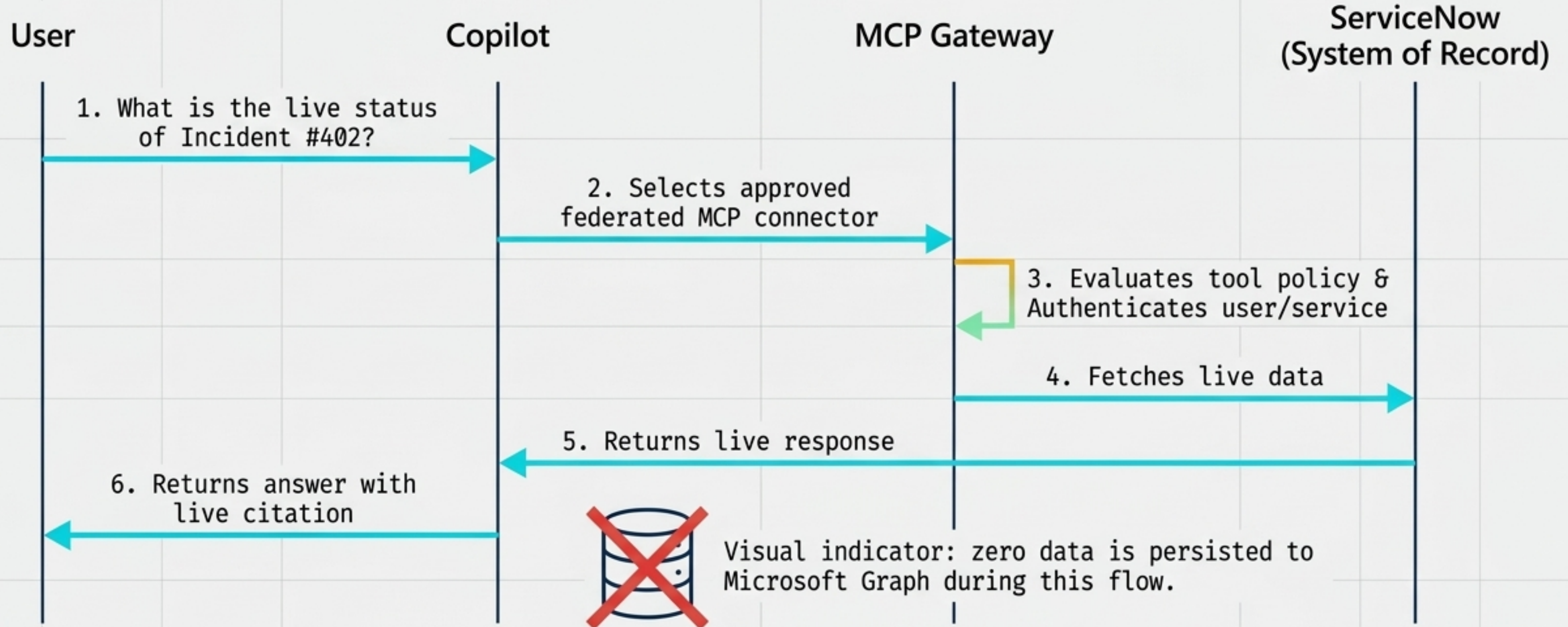
The Knowledge Plane: Building Synced Connectors



PERMISSION FIDELITY IS NON-NEGOTIABLE.

If a connector indexes source text but fails to perfectly model the source authorization, AI amplifies an existing data exposure risk.

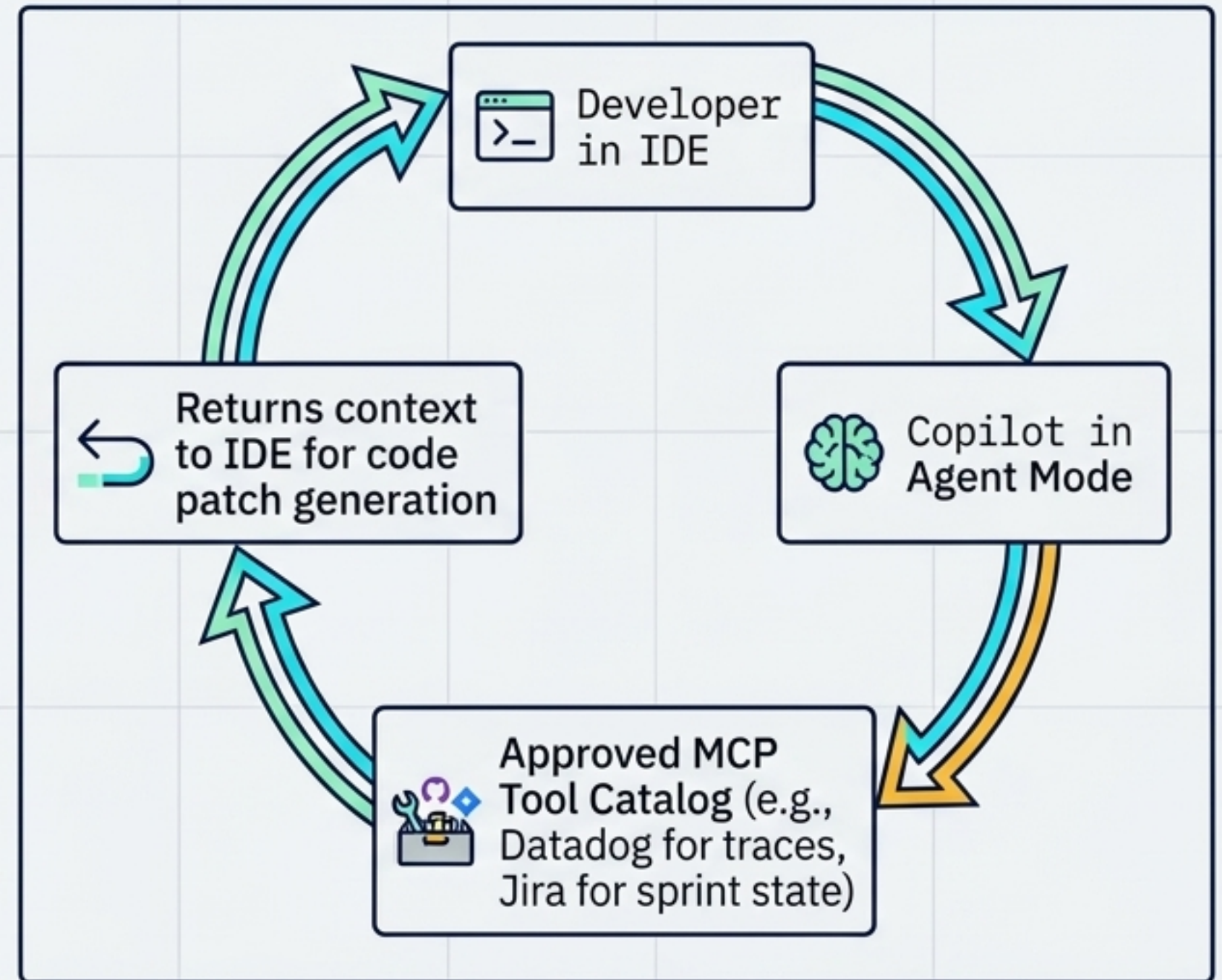
The Transaction Plane: Federated Live Fetch (MCP)



Why MCP? Essential for rapid data changes, enforcing regulatory boundaries, and executing governed write-actions (e.g., ticket creation) without duplicating records.

GitHub Extensibility: Identity and Tool Access

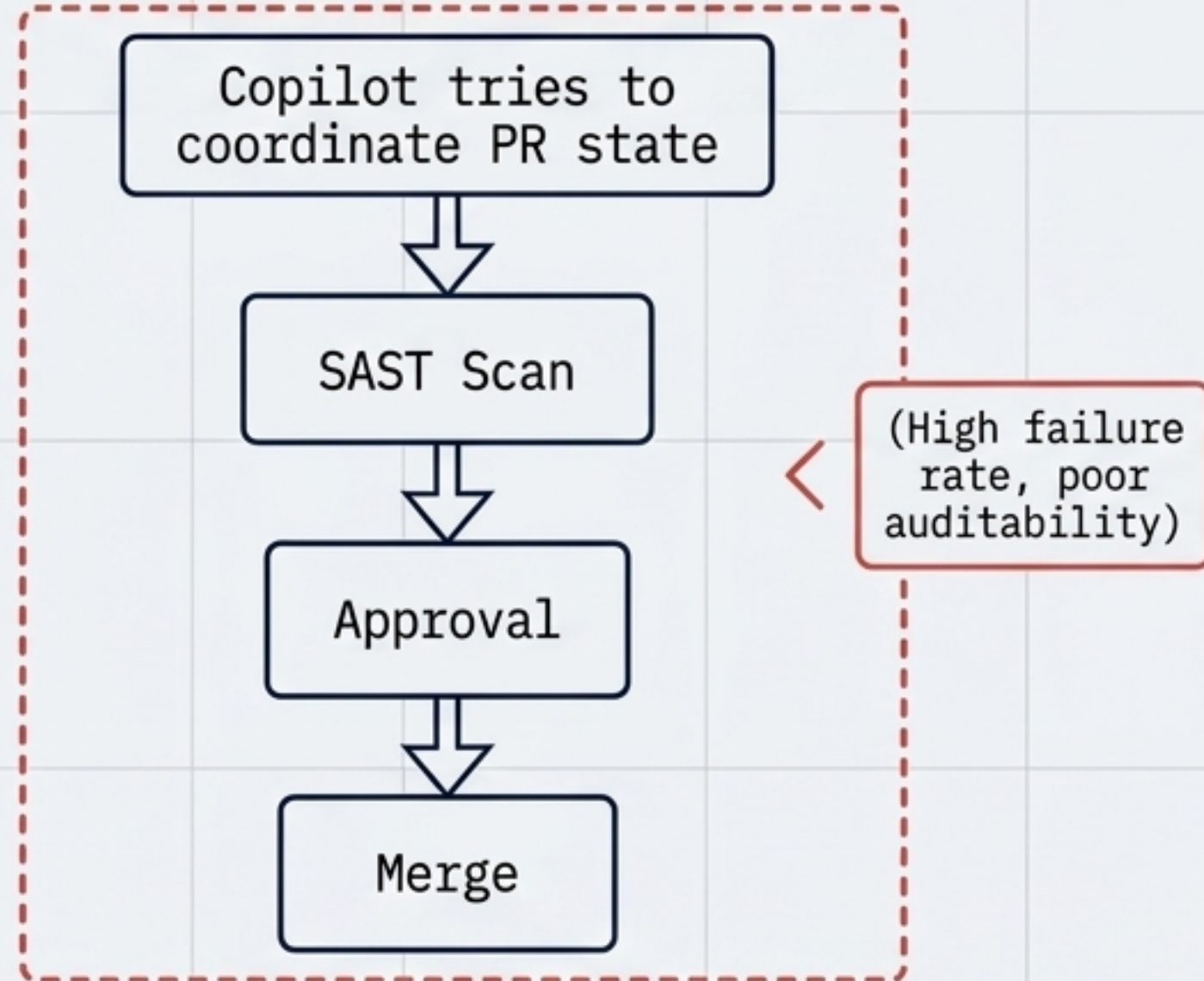
Identity Type	Usage Recommendation
GitHub App	Default for server-to-server, fine-grained, short-lived tokens. ✓
OAuth App	Strictly for user-delegated workflows.
PAT	Avoid for durable enterprise integrations. ⚠



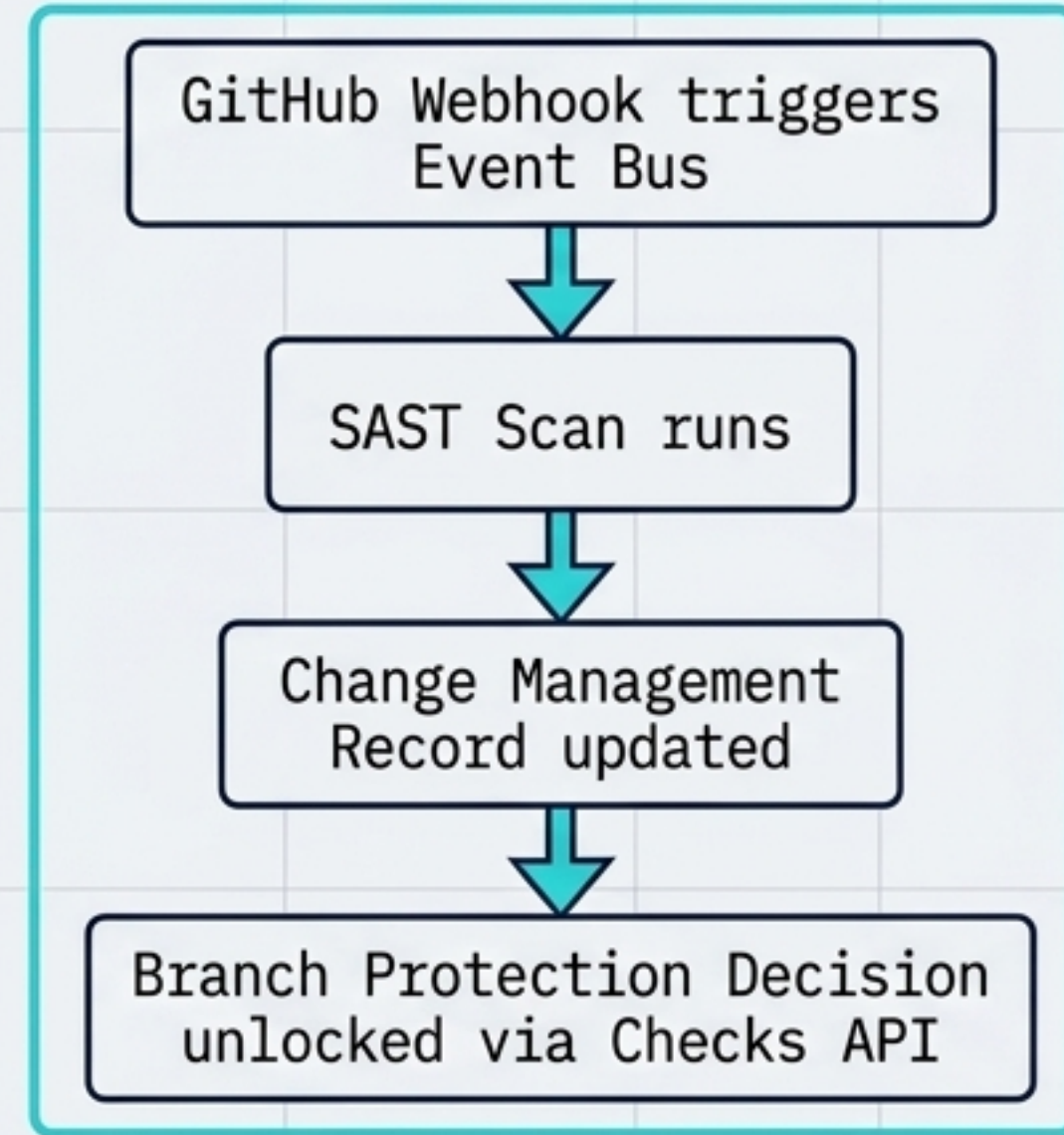
Strategic Shift Note: GitHub Copilot Extensions are deprecated. Model Context Protocol (MCP) servers are now the singular standard for third-party AI tool integration.

State Transitions: APIs Govern, AI Accelerates

The Fragile Way



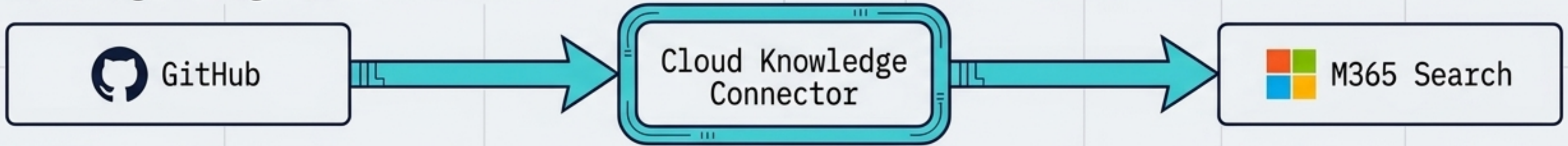
The Durable Way



RULE: DO NOT RELY ON LLMs FOR LIFECYCLE STATE COORDINATION. AI can synthesize, generate, and accelerate. Event integrations (Webhooks, Checks API) must remain the unyielding system of record for state transitions.

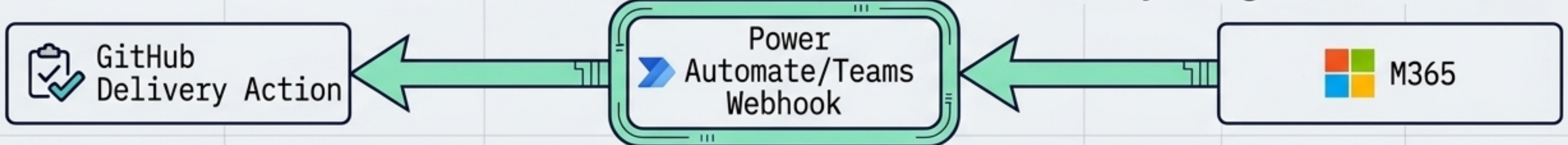
Bridging the Domains: Cross-Platform Patterns

Knowledge Bridge (GitHub -> M365)



Use Case: A PM queries engineering specs (ADRs and Markdown) from Microsoft Teams.

Delivery Bridge (M365 -> GitHub)

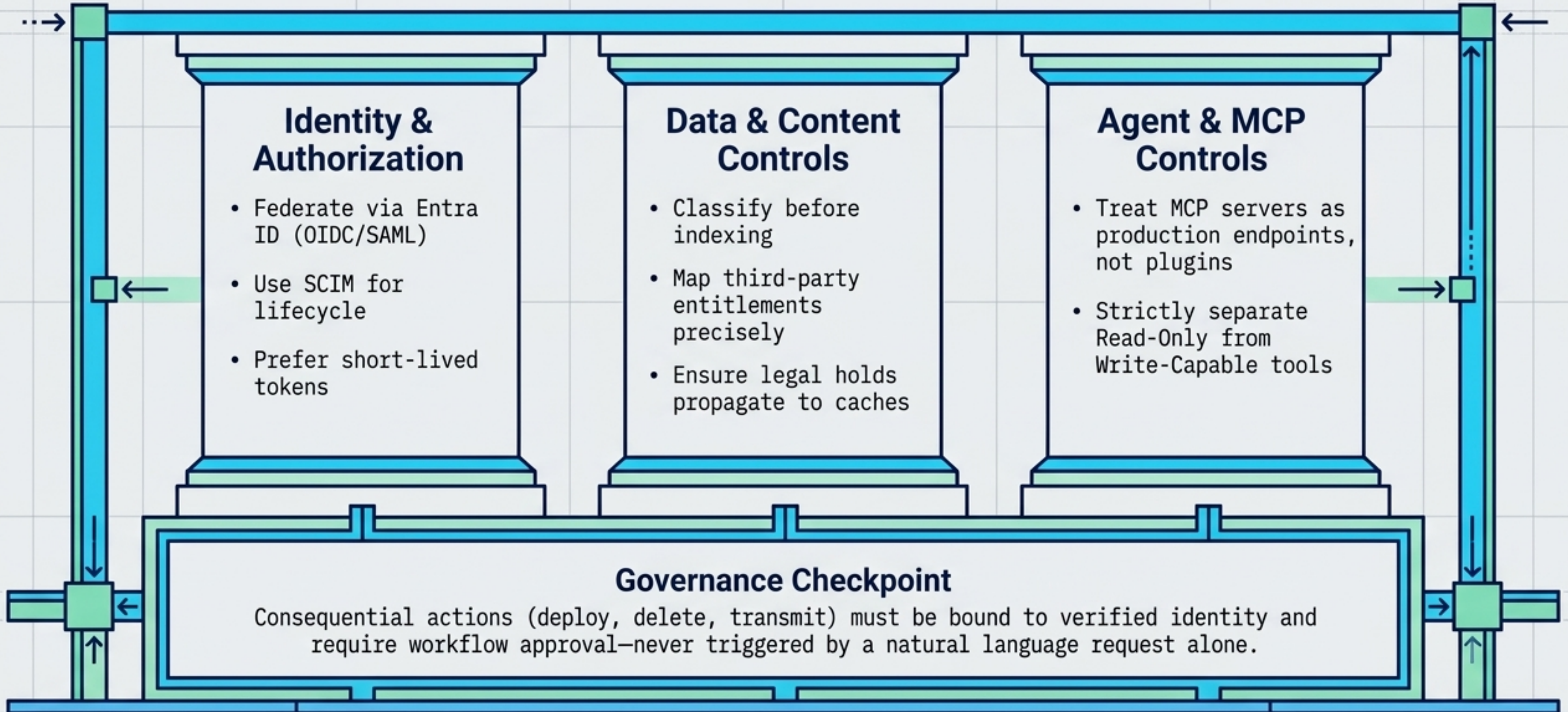


Use Case: A PR approval triggers a ServiceNow control step, mirrored back to GitHub Checks.

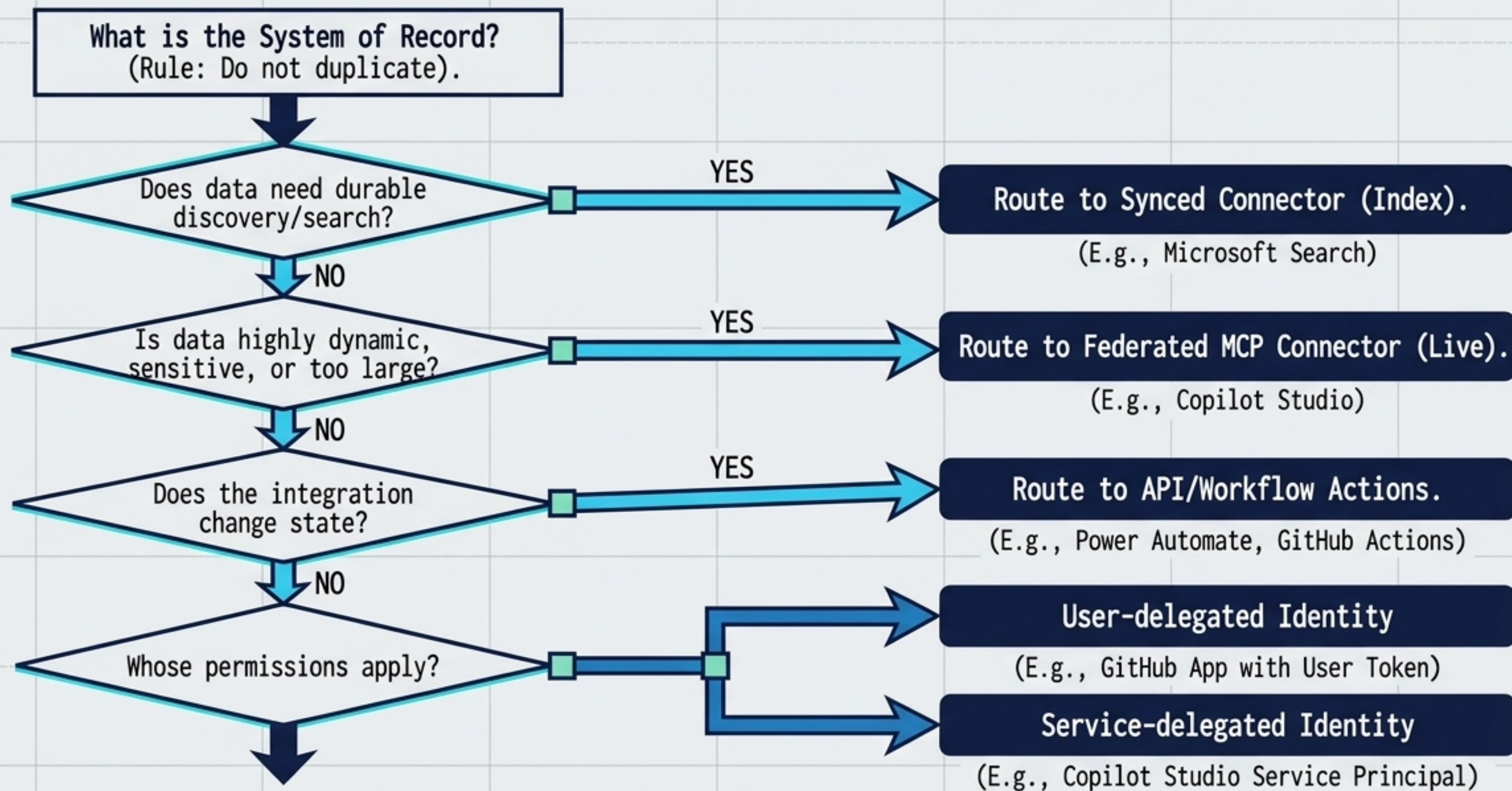
Design Principle

Preserve source-of-truth boundaries. Office documents live in SharePoint; Source code and Developer evidence live in GitHub. Do not cross-contaminate the systems of record.

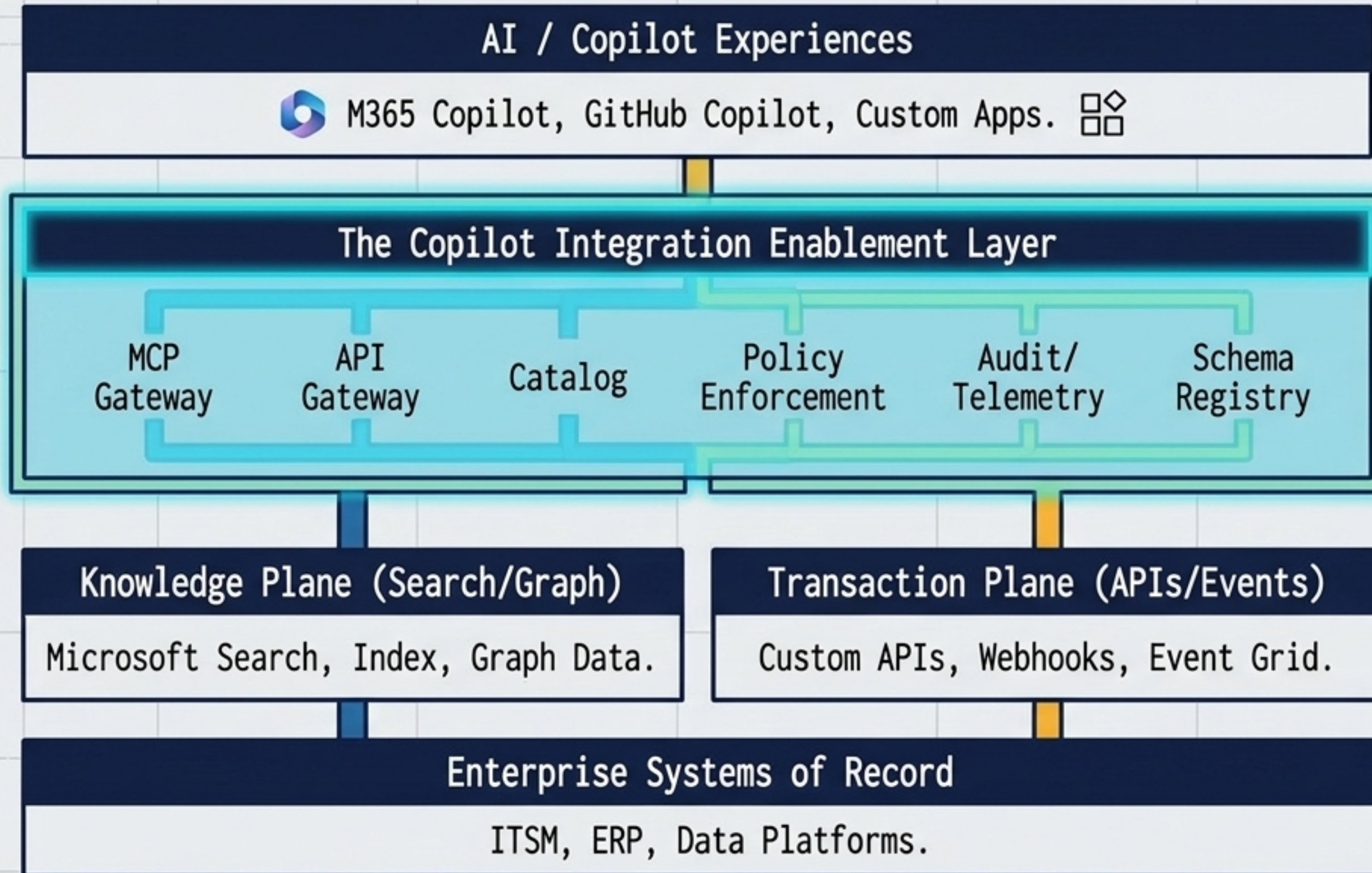
Security Architecture & Trust Boundaries



The Integration Decision Framework



The Target State: A Unified AI Extensibility Stack



Synthesis Insight

This layered approach structurally eliminates shadow point-to-point AI connections. It centralizes governance while decentralizing AI productivity.

Strategic Imperative: Govern the Extensibility Layer

The Ask

Mandate a centralized “Copilot Integration Enablement Layer” to govern all MCP and connector traffic across M365 and GitHub.

The Stakes

Doing nothing shatters permission fidelity and invites ungoverned shadow AI sprawl. Acting unifies AI auditability and capability.

The Golden Rule

Never duplicate a system of record just to make it “AI accessible.”

Immediate Next Step

Audit all existing GitHub OAuth apps and transition critical server-to-server workflows to GitHub Apps and MCP.